Machine Learning for Classifying Top 10% Most-Viewed BBC News YouTube Videos

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Abstract

Classifying highly viewed online videos is an important task for media organizations and content creators because it helps identify the factors associated with strong audience attention. Machine learning models can analyze video metadata and interaction-related variables to distinguish highly successful videos from less successful ones.

This study develops machine learning models to classify whether a video published by BBC News belongs to the **top 10% most-viewed videos** in the dataset. The target variable is defined using the **90th percentile of view counts**, corresponding to **469,012 views**. Videos with view counts greater than or equal to this threshold are labeled as top-performing videos, while the remaining videos are labeled as non-top-performing videos.

To provide a more meaningful evaluation, two feature settings are compared. The first uses only metadata and title-related features, while the second additionally incorporates interaction-based variables such as likes, comments, engagement, and the like-comment ratio. Three classification models are evaluated: Logistic Regression, Random Forest, and XGBoost.

The results show that the **full-feature setting** substantially outperforms the metadata-only setting. Under the full-feature setting, **Random Forest achieved the best overall performance**, with an accuracy of **0.9640**, precision of **0.8422**, recall of **0.7875**, and F1-score of **0.8140**. Under the metadata-only setting, performance was much lower, although XGBoost achieved the highest F1-score

of **0.2328**. Feature importance analysis further indicates that interaction-based variables, especially **likeCount** and **engagement**, are the strongest contributors to classification performance.

A simple Streamlit-based interface is also implemented to demonstrate practical application. Overall, the study shows that machine learning can effectively classify top-performing BBC News videos, while also highlighting the importance of feature selection and careful interpretation of interaction-based variables.

Keywords: YouTube video classification, BBC News, machine learning, Random Forest, XGBoost, top 10% views

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1. Introduction

Online video platforms generate large volumes of data related to audience behavior and content performance. Variables such as view count, likes, comments, upload time, and title characteristics provide useful signals for understanding how viewers respond to content. For media organizations, analyzing these signals can help explain which videos attract stronger public attention and why.

Machine learning has been widely applied to structured datasets in order to identify patterns and support predictive or classificatory decision-making. In the context of YouTube, machine learning models can be trained to distinguish highly successful videos from less successful ones based on metadata and interaction-related features. Such analysis may help content publishers better understand content performance and improve future strategy.

This study focuses on videos published by **BBC News** and aims to classify whether a video belongs to the **top 10% most-viewed videos** in the dataset. Rather than using an arbitrary fixed threshold, the target variable is defined using the **90th percentile of view counts**, which provides a more data-driven way to identify highly viewed content.

To make the analysis more academically meaningful, the study compares two feature settings. The first uses only metadata and title-related variables, which are closer to an early-stage prediction setting. The second includes interaction-based variables, which are more strongly associated with actual performance. This comparison helps distinguish between limited early-stage classification and stronger post-publication classification based on audience response.

In addition, the study compares three machine learning models—Logistic Regression, Random Forest, and XGBoost—and examines which variables contribute most strongly to the final classification outcome. A simple Streamlit-based system is also implemented to demonstrate how the trained model can be used in a practical setting.

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2. Background and Related Work

The problem of analyzing online content popularity has received significant attention in machine learning and social media research. Prior studies have shown that video performance is influenced by multiple factors, including content metadata, upload timing, textual characteristics, and audience

interaction signals. In many cases, interaction-based variables such as likes and comments are strongly associated with popularity because they reflect how viewers respond to content after publication.

For structured tabular datasets, common machine learning models include Logistic Regression, Random Forest, and gradient boosting methods such as XGBoost. Logistic Regression is often used as a baseline due to its simplicity and interpretability. Random Forest is effective for capturing non-linear relationships through ensemble learning, while XGBoost is widely recognized for strong predictive performance on tabular data.

This study differs from broader popularity prediction work in three ways. First, it focuses specifically on **BBC News**, which provides a consistent domain of news-related YouTube videos. Second, it defines high performance using the **top 10% of view counts**, making the target variable relative to the dataset distribution rather than dependent on a fixed absolute threshold. Third, it compares **metadata-only** and **full-feature** settings to better distinguish early-stage signals from post-publication interaction signals.

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3. Dataset and Target Definition

3.1 Dataset

The dataset contains **20,000 YouTube videos** published by **BBC News**. Each record includes information such as video title, publishing time, duration, view count, like count, and comment count. These raw variables are then processed to create a structured feature set for machine learning.

3.2 Target Variable

The target variable is defined using the **90th percentile of view counts**. In this dataset, the 90th percentile corresponds to **469,012 views**. Videos with view counts greater than or equal to **469,012** are labeled as **1**, representing videos in the **top 10% most-viewed group**. The remaining videos are labeled as **0**.

This formulation creates a binary classification task and reflects relative popularity within the dataset more effectively than a manually chosen fixed threshold.

3.3 Class Distribution

The resulting class distribution is imbalanced:

- **Class 0 (not top 10%)**: 18,000 videos
- **Class 1 (top 10%)**: 2,000 videos

This corresponds to a **90% / 10%** split, which makes it necessary to evaluate model performance using not only accuracy but also precision, recall, and F1-score.

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4. Methodology

4.1 Data Preprocessing

Before model training, several preprocessing steps are performed to improve data quality and feature usability. Missing or inconsistent values are checked and handled where necessary. Video duration is converted into seconds so that it can be used as a numerical feature.

Publishing timestamps are also processed to derive temporal variables such as upload hour and upload day. The upload hour is encoded using sine and cosine transformations in order to better represent the cyclical nature of time. Day-of-week information is transformed into indicator variables for model input.

4.2 Feature Engineering

To support comparison between different information sources, the study uses two feature settings.

Metadata-only features

The metadata-only setting includes variables that are available at or near publication time:

- duration_seconds
- title_length
- title_word_count
- has_keyword
- has_number_in_title
- has_question_mark
- hour_sin
- hour_cos
- is_weekend
- upload day indicator variables

These variables are intended to support a more realistic early-stage classification setting.

Full-feature setting

The full-feature setting includes all metadata variables as well as interaction-based variables:

- likeCount
- commentCount
- engagement (likes + comments)
- like_comment_ratio

These features are expected to improve classification performance because they capture audience response more directly.

4.3 Experimental Settings

Two experiments are conducted:

Experiment 1: Metadata-only setting
This setting uses only metadata, title-related, and temporal variables. It represents a more constrained and more realistic early-stage classification scenario.

Experiment 2: Full-feature setting
This setting includes both metadata features and interaction-based variables. It represents classification using stronger post-publication signals.

4.4 Machine Learning Models

Three models are used in both settings:

- **Logistic Regression**
- **Random Forest**
- **XGBoost**

The dataset is split into training and testing subsets using a stratified sampling strategy in order to preserve the class distribution. Model performance is then evaluated on the test set.

4.5 Evaluation Metrics

Because the dataset is imbalanced, model performance is assessed using four metrics:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**

Among these metrics, **F1-score** is especially important because it balances precision and recall and provides a more reliable measure of performance under class imbalance.

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5. Results

The two feature settings produced substantially different outcomes. The metadata-only setting delivered limited performance, whereas the full-feature setting greatly improved classification ability.

Table 1. Model Performance under Different Feature Settings

Feature Setting	Model	Accuracy	Precision	Recall	F1-score
Metadata-only	Logistic Regression	0.5962	0.1412	0.5975	0.2284
Metadata-only	Random Forest	0.8980	0.3333	0.0200	0.0377
Metadata-only	XGBoost	0.7298	0.1625	0.4100	0.2328
Full-feature	Logistic Regression	0.9350	0.6159	0.9300	0.7410
Full-feature	Random Forest	0.9640	0.8422	0.7875	0.8140
Full-feature	XGBoost	0.9553	0.7241	0.8925	0.7996

Under the **metadata-only setting**, model performance was relatively weak. Logistic Regression achieved the highest recall (**0.5975**) but very low precision (**0.1412**), indicating that it identified many positive cases at the cost of many false positives. Random Forest showed high accuracy (**0.8980**), but its recall was only **0.0200**, meaning that it failed to identify most top 10% videos. XGBoost achieved the highest F1-score (**0.2328**) in this setting, providing the best overall balance between precision and recall.

Under the **full-feature setting**, all models improved substantially. Logistic Regression achieved a strong recall of **0.9300**, indicating high sensitivity to top-performing videos. However, **Random Forest achieved the best overall performance**, with the highest accuracy (**0.9640**), precision (**0.8422**), and F1-score (**0.8140**). XGBoost also performed strongly, especially in recall (**0.8925**) and F1-score (**0.7996**), but remained slightly below Random Forest overall.

These results show that interaction-based variables dramatically improve classification performance. While metadata-only features provide limited predictive information, the addition of likes, comments, engagement, and the like-comment ratio makes the models much more effective at identifying highly viewed videos.

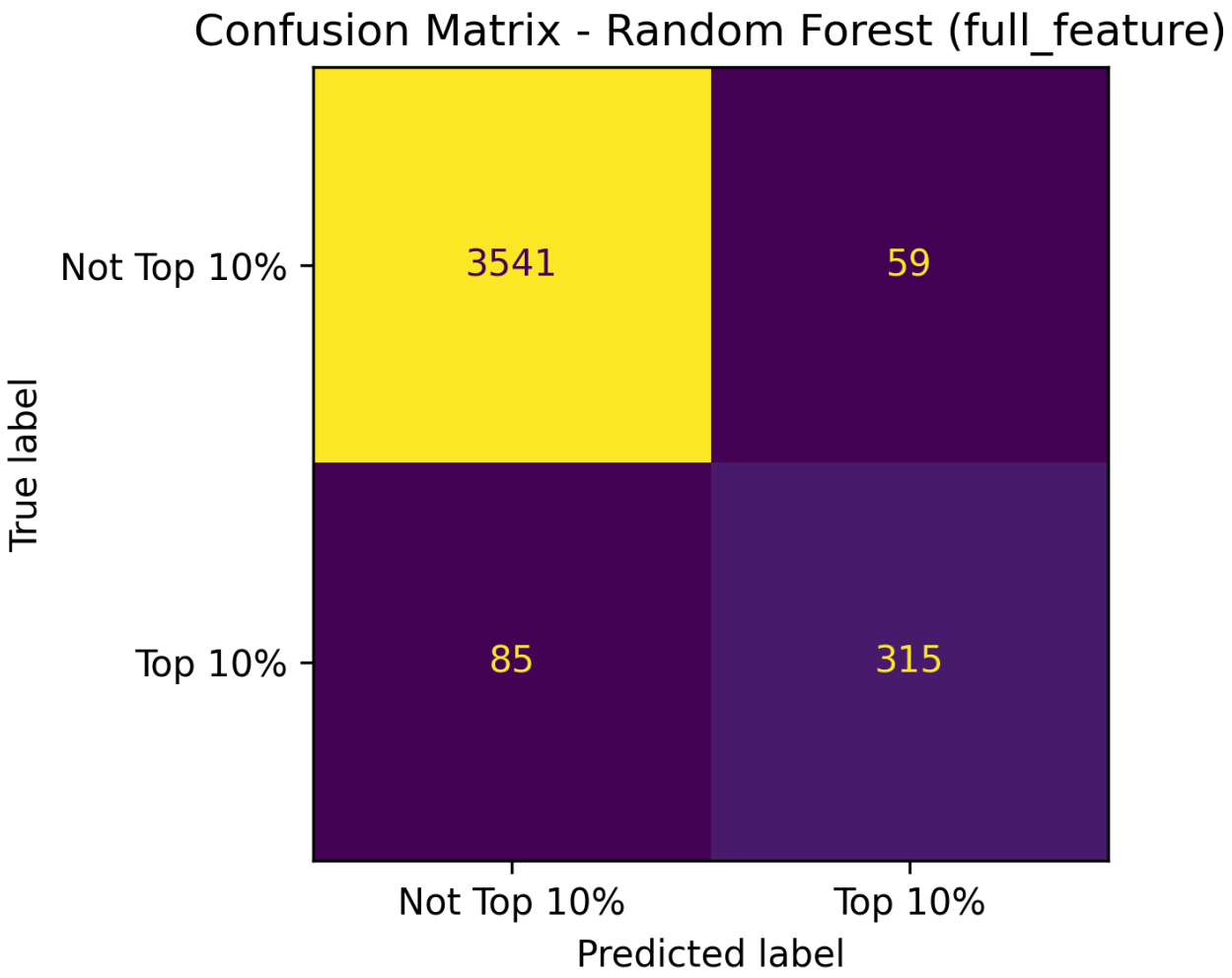

Figure 1: *Confusion matrix of the Random Forest model under the full-feature setting.*

The confusion matrix of the best-performing model provides a clearer view of the classification outcome and shows that the model can separate the two classes effectively, while still making some mistakes in identifying minority-class videos.

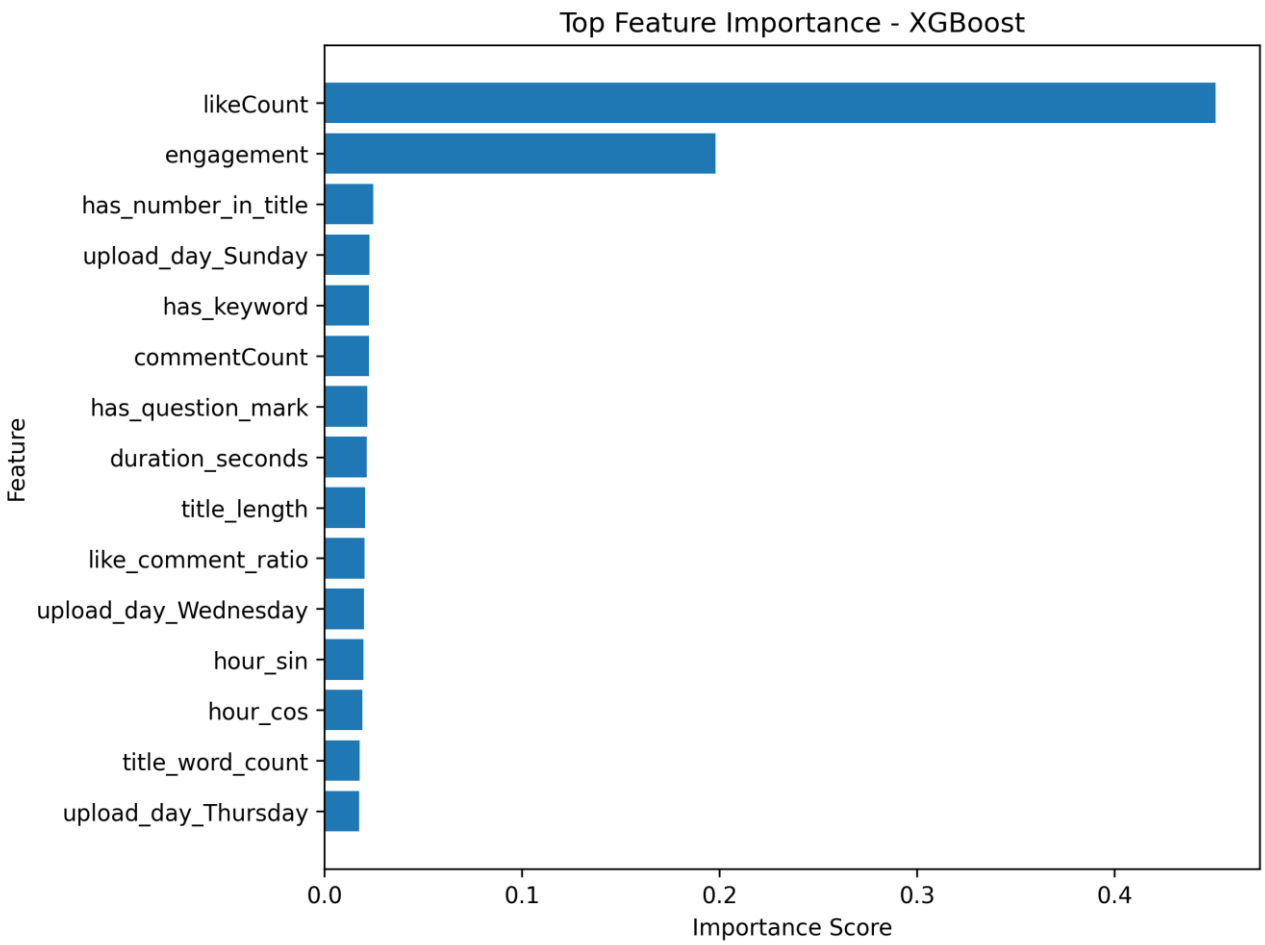

Figure 2: Top 15 feature importance scores of the XGBoost model under the full-feature setting.

Feature importance analysis indicates that **likeCount** is the most influential feature, followed by **engagement**. Other useful features include **has_number_in_title**, **upload_day_Sunday**, **has_keyword**, **commentCount**, **has_question_mark**, and **duration_seconds**. This suggests that interaction-based variables dominate the classification process, while title-related and temporal features provide supporting information.

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6. System Implementation

To demonstrate practical application, a simple web-based prototype is implemented using the **Streamlit** framework. The system allows users to enter video-related input variables and obtain a classification result indicating whether the video is likely to belong to the **top 10% most-viewed group**.

The interface accepts structured inputs such as video duration, title characteristics, upload time, and interaction-related variables. Once the user submits the input values, the trained model processes the feature set and returns the predicted class. This allows users to interact with the model directly and observe how input variables may influence the classification result.

The Streamlit prototype helps bridge the gap between model development and practical application. It also demonstrates how machine learning models can be integrated into a simple decision-support tool for media analysis.

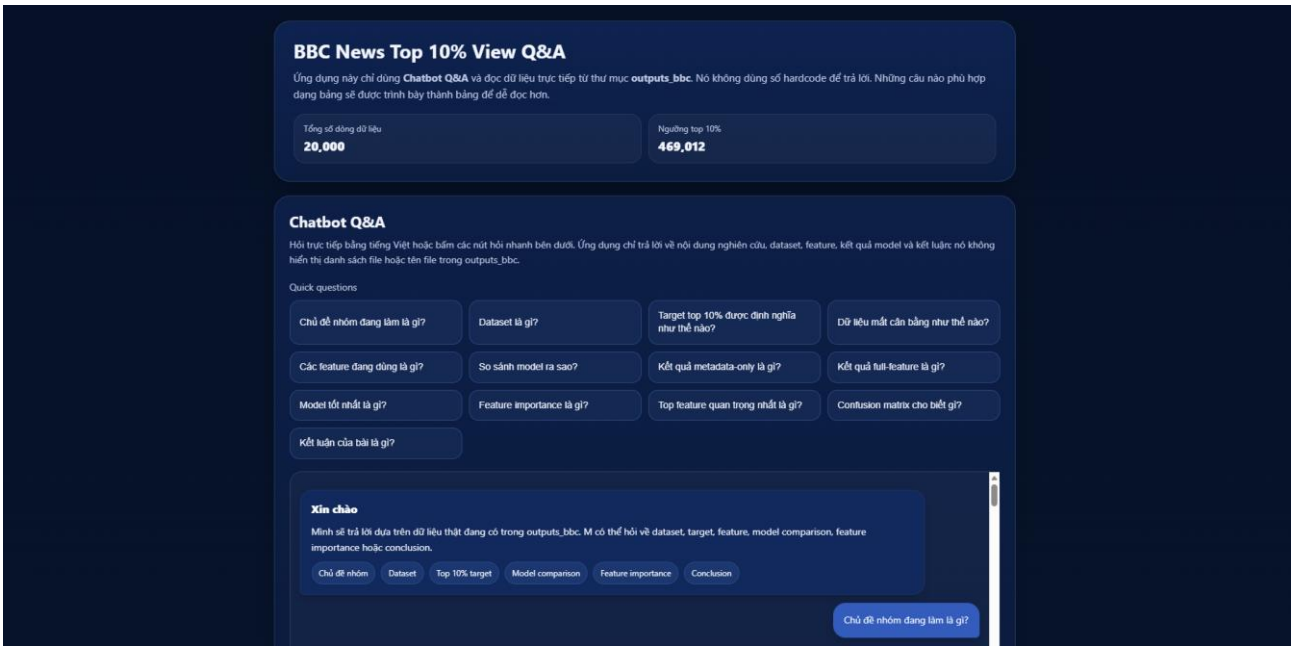

Figure 3: *Screenshot of the Streamlit interface*

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7. Discussion

The results reveal a clear distinction between the two experimental settings. In the **metadata-only setting**, all models performed substantially worse than in the full-feature setting, indicating that metadata and title-related variables alone provide limited power for identifying whether a video belongs to the top 10% most-viewed group.

Among the metadata-only models, **XGBoost achieved the highest F1-score (0.2328)**, suggesting the best overall balance between precision and recall. Logistic Regression achieved the highest recall (**0.5975**), meaning it captured more top-performing videos, but it also produced many false positives. Random Forest, despite its high accuracy, performed poorly in this setting because its recall was extremely low (**0.0200**). This result demonstrates why accuracy alone is not sufficient for evaluating performance on imbalanced data.

In the **full-feature setting**, model performance improved dramatically. This confirms that interaction-based variables such as likes, comments, and engagement are strongly associated with whether a video becomes one of the most-viewed items in the dataset. **Random Forest achieved the strongest**

overall results, while XGBoost also showed strong and balanced performance. Logistic Regression remained highly competitive in recall, which suggests that it was effective at identifying positive cases, although with more false positives than the tree-based models.

The feature importance analysis further supports these findings. Variables such as **likeCount** and **engagement** contributed most strongly to the classification process, while metadata and title-based features played a secondary role. This indicates that audience response is a much stronger signal of video success than publication-time metadata alone.

However, this finding must be interpreted carefully. The strong performance of the full-feature models does not necessarily imply that the system can forecast long-term success before a video is published. Instead, it shows that once interaction-related signals become available, machine learning models can classify highly viewed videos much more effectively. Therefore, the metadata-only setting is more appropriate for limited early-stage classification, while the full-feature setting is better understood as post-publication classification.

This study also has several limitations. First, the dataset includes only **BBC News** videos, so the findings may not generalize equally well to other channels or content categories. Second, the title-based features are relatively simple and do not capture deeper semantic information. Future work could incorporate more advanced natural language processing features, test additional machine learning models, and expand the dataset across multiple channels.

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8. Conclusion

This study presented a machine learning approach for classifying whether a BBC News video belongs to the **top 10% most-viewed videos** in the dataset. The target variable was defined using the **90th percentile of view counts**, corresponding to **469,012 views**.

Two feature settings were compared: a **metadata-only setting** and a **full-feature setting** that included interaction-based variables. The results showed that metadata-only models had limited performance, although XGBoost achieved the highest F1-score (**0.2328**) in that setting. In contrast, the full-feature models performed substantially better, highlighting the importance of audience interaction signals.

Among all evaluated models, **Random Forest achieved the best overall performance in the full-feature setting**, with an accuracy of **0.9640**, precision of **0.8422**, recall of **0.7875**, and F1-score of **0.8140**. Feature importance analysis also indicated that **likeCount** and **engagement** were the most influential variables in the classification process.

Overall, the study shows that machine learning can effectively classify highly viewed BBC News videos using structured YouTube data. At the same time, it highlights the importance of feature selection, class imbalance awareness, and careful interpretation of interaction-based variables when evaluating video performance.

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